

CARISURG MEDTECH PATHWAYS

Digital Human-In-The-Loop Decision Support For Triage at Mercer General Emergency Department

Research Proposal



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LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
AUC	Area Under the Curve
DPIA	Data Protection Impact Assessment
ED	Emergency Department
ESI	Emergency Severity Index
ESM	Epic Sepsis Model
HCP	Healthcare Professional
IT	Information Technology
LLM	Large Language Model
MEWS	Modified Early Warning Score
NASA-TLX	National Aeronautics and Space Administration Task Load Index
NASSS	Non-adoption, Abandonment, Scale-up, Spread, and Sustainability
WHO	World Health Organisation

1. STATEMENT OF THE PROBLEM

Mercer General Emergency Department triages 70 to 170 patients daily using paper-based Emergency Severity Index (ESI) forms, requiring nurses to make rapid acuity decisions under high workload with no structured digital support (1). Existing AI triage tools optimise predictive accuracy but rarely support structured information gathering in low-resource settings (2,3). This proposal tests whether a human-in-the-loop digital triage tool, piloted over a 12-week period, reduces missing ESI fields and improves alignment with rule-based classifications compared with the existing paper workflow. The hypothesis is rejected if no measurable improvement is observed.

2. PREVIOUS WORK

2.1 AI in Clinical Prediction

The emergency department (ED) is consistently described as a high-pressure environment shaped by overcrowding, staffing shortages, and variability in triage decision-making. Across multiple healthcare systems, these pressures contribute to delays in patient assessment and increased risk of inconsistent prioritisation (3,4). In Caribbean ED contexts, these constraints are further intensified by limited staffing, constrained diagnostic access, and structural bottlenecks in patient flow, including non-standardised movement between care stages (2). Taken together, these findings indicate that ED triage is not only a classification task, but a time-sensitive operational decision made under sustained system pressure.

Within this environment, triage is further complicated by cognitive burden at the point of decision-making. Cognitive task analysis of emergency nurses shows that triage decisions are made under conditions of incomplete information, frequent interruptions, and fatigue, all of which increase the likelihood of variability and error (1). Gorick et al. (5) extend this finding by demonstrating that while less experienced nurses rely on structured frameworks, more experienced nurses layer intuition, visual cues, and contextualised vital sign interpretation, yet both groups described environments where pressure to maintain patient flow frequently overrode time for thorough assessment. This suggests that triage performance is not determined solely by clinical knowledge, but also by the ability to structure and process fragmented patient information under time constraints. However, most existing digital interventions focus on system efficiency rather than explicitly supporting the cognitive processes involved in triage decision-making.

In parallel, machine learning approaches to ED triage have demonstrated that predictive models can approximate clinician-level decisions in controlled retrospective settings, but their performance is often limited by generalisability and reliance on structured historical data (3,4). More recent large language model (LLM)-based studies extend this work by enabling free-text reasoning over clinical scenarios. Nedos et al. (6) report moderate agreement between LLM-generated triage predictions and physician-assigned ESI scores across a large retrospective dataset, while Wang et al. (7) similarly demonstrate moderate performance in ED triage and referral tasks across multiple metrics. Araouchi and Adda (8) propose a multimodal AI-assisted triage system combining multiple machine learning models, including stacking approaches, and report that ensemble methods improve predictive performance compared with individual classifiers in retrospective ED datasets. However, both the LLM-based studies and the multimodal ML systems highlight variability across tasks and do not evaluate integration into real-time clinical workflows.

Across these AI approaches, a consistent limitation emerges: systems are typically evaluated as standalone prediction tools rather than as components of a triage workflow. They focus primarily on output accuracy, while providing limited support for structured information gathering or clinician decision-making under uncertainty. As a result, there remains a disconnect between algorithmic prediction performance and the practical cognitive demands of emergency triage.

2.2 Ethics, Safety, and Governance

Recent evidence from deployed clinical AI systems demonstrates that this disconnect carries patient safety consequences that extend beyond accuracy metrics alone. The World Health Organisation (9) established six consensus principles for ethical AI in health, including transparency, accountability, and equity, and explicitly states that these must be embedded at the design stage rather than applied retrospectively. This normative framework is directly relevant to triage AI deployment: it identifies equity and inclusivity as foundational requirements, warning that AI systems trained on data that underrepresents certain populations risk systematically disadvantaging already-marginalised communities.

The mechanism by which this occurs is not always deliberate. Obermeyer et al. (10), in a landmark study published in *Science*, demonstrated that a widely deployed commercial health risk-prediction algorithm systematically under-prioritised Black patients, not because of explicit discriminatory intent, but because it used predicted healthcare cost as a proxy for health need.

Since Black patients historically incurred lower costs at equivalent levels of illness severity due to structural barriers to access, the algorithm learned to rank them as less unwell. The finding is directly transferable to emergency triage contexts: variables that appear clinically neutral, including prior visit count, mode of arrival, or time to first assessment, may function as proxies for socioeconomic disadvantage or differential access, encoding existing inequality into an automated recommendation. Every proxy variable in a triage support model therefore requires interrogation, not only for what it measures, but for what it correlates with that was never intended.

Real-world deployment evidence confirms that these are not theoretical concerns. Wong et al. (11) conducted an independent external validation of the Epic Sepsis Model (ESM), a proprietary AI tool deployed across hundreds of hospitals in the United States and found that the model missed two-thirds of sepsis cases and generated alerts requiring clinicians to review 109 cases to identify a single true positive. A subsequent study by Kamran et al. (12) established that the ESM's apparent predictive performance was partially an artefact of data leakage: the model had been trained on variables that only appear in patient records after a clinician has already begun treating suspected sepsis, making it reactive rather than predictive. The ESM case illustrates precisely what the WHO (9) framework warns against: deployment at scale in the absence of independent validation, transparent governance, or post-market accountability mechanisms.

Taken together, these findings illustrate two categories of failure directly relevant to nurse-facing triage support tools: technical failures in model design and validation, including distribution shift and circular reasoning from data leakage, and human-factors failures, including alert fatigue driven by high false-positive rates that erode clinician vigilance over time. Hogg et al. (13) contextualise these risks within the broader implementation literature, demonstrating through a synthesis of 111 studies that clinical AI adoption is frequently undermined by clinician scepticism about AI decision-making and inadequate post-deployment monitoring, particularly in resource-constrained settings underrepresented in training data. These failure modes, specifically high false-positive alert rates eroding nurse vigilance and silent model performance decay as case-mix evolves, are formally captured as high-priority risks in the project risk register (Table 2, Section 6) and directly inform the mitigation design of the proposed system.

2.3 Gaps in the Literature

The literature therefore suggests four partially addressed domains: operational ED pressure, cognitive load in triage decision-making, predictive AI performance in clinical classification tasks,

and the safety and governance risks that emerge when AI tools are deployed without adequate validation or human-factors consideration. These domains are not yet integrated into a single system that simultaneously supports structured data collection, reinforces clinical reasoning, maintains rule-based safety constraints, and operates within a transparent governance framework appropriate to resource-limited Caribbean ED settings. This fragmentation indicates a design gap rather than a lack of individual technological solutions. What is needed is not a more accurate prediction model, but a lightweight, human-in-the-loop system designed from the ground up for the operational, equity, and governance constraints of a Caribbean emergency department: one that keeps the nurse as the decision-maker, makes its reasoning transparent and auditable, and does not introduce the proxy-variable risks or alert-fatigue dynamics that have undermined deployed clinical AI elsewhere.

3. PROPOSED SOLUTION

3.1 Overview

The proposed system is a nurse-facing digital triage support tool designed to convert the current paper-based Emergency Severity Index (ESI) workflow into a structured electronic process. It is intended as a clinical decision-support system that enhances documentation quality, completeness, and consistency without replacing or reducing nurse clinical judgement. The system operates at the point of patient intake and is used by triage nurses via a touchscreen interface. It is explicitly designed to support, not automate, triage decision-making.

3.2 System Design

As shown in Figure 1, the system expands upon the initial draft layout and consists of five integrated components that map directly onto the clinical pathway.

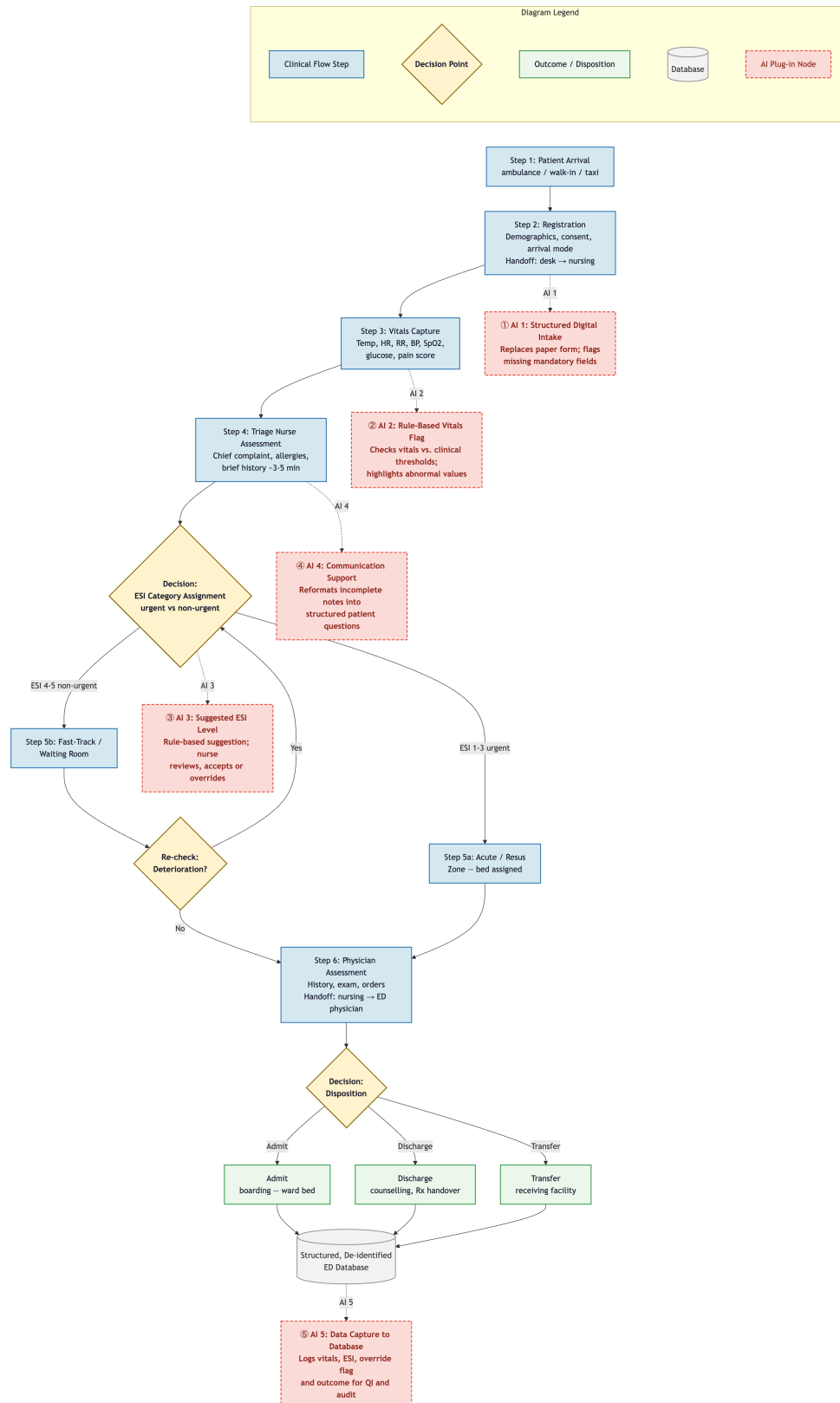


Figure 1. Door-to-disposition workflow at Mercer General ED, annotated with five AI plug-in points (AI 1-AI 5).

Component 1: Structured Digital Intake (AI 1)

A nurse-facing digital interface replaces the paper ESI front sheet and standardises data entry during registration. It enforces the consistent capture of patient demographics, chief complaints, arrival modes, and consent. The system actively identifies missing or incomplete mandatory data fields and prevents progression until the entry omissions are resolved.

Component 2: Rule-Based Vitals Flag (AI 2)

A deterministic, clinical rules engine processes structured data immediately following vital signs capture. It automatically checks recorded parameters (temperature, heart rate, respiratory rate, blood pressure, oxygen saturation, blood glucose, and pain score) against predefined clinical thresholds. Any abnormal values are highlighted instantly on the screen to manage cognitive load before acuity assignment.

Component 3: Suggested ESI Level (AI 3)

Integrating the formal 5-level ESI protocol, this component uses fixed, transparent clinical rules to generate an automated triage level suggestion. This component is fully rule-based and does not rely on probabilistic machine learning. The tool supports the decision but does not make it; the triage nurse must actively review, accept, or override the suggestion with a logged reason. This design directly mitigates the risk of automation bias, a named high-priority ethical risk in the project register (Table 2, Row 4), whereby clinicians may unconsciously defer to AI output without independent reassessment.

Component 4: AI-Assisted Communication Support Module (AI 4)

An integrated language module supports nurses in clarifying ambiguous or rough free-text triage notes. The system connects to an API to reframe incomplete notes into structured, clear, and polite patient-facing questions that the nurse can ask to fill information gaps. This component is strictly limited to language reformatting and does not perform clinical diagnosis or risk stratification.

Component 5: Structured Data Capture to Database (AI 5)

At the final point of patient disposition, the system automatically logs the entire structured encounter, including initial vitals, the generated ESI suggestion, nurse override flags, and the outcome. This data is transferred into a de-identified, queryable ED database, creating the digital foundation required for retrospective quality-improvement loops and operational audit trails.

3.3 Evaluation Plan

To protect patient safety, the system will not be tested live on patients during this phase. The 12-week pilot instead combines two strands of evidence. First, the rule-based components (AI 1–AI 3, AI 5) will be evaluated in a structured simulation against synthetic Mercer-representative triage cases built from the Appendix A front-sheet fields, scored for field completeness and ESI agreement against the existing rule set. Second, the Hong et al. (14) dataset of 560,486 de-identified US ED visits is used only as a feasibility benchmark for the underlying classifier architecture, not as a validation source for Mercer-specific ESI accuracy, because it predicts hospital admission rather than ESI level and its population (Yale New Haven, US) does not match Mercer's demographic or disease profile.

Demographic limitation: The Hong et al. dataset originates from a large US urban hospital and cannot stand in for Caribbean demographic, disease-pattern, or staffing realities. No claim of clinical validity is made from this benchmark alone; full prospective validation on Mercer-representative data is a precondition for any live deployment beyond this 12-week pilot.

Objective: Demonstrate that structured digital data gathering reduces missing ESI fields and improves alignment with rule-based ESI logic compared with the existing paper workflow, within the 12-week pilot window. The hypothesis is rejected if no measurable improvement in either metric is observed.

3.4 Objectives

The overall aim of the system is to improve the quality, safety, and consistency of triage documentation in a resource-constrained emergency department environment.

Primary objective

- Improve completeness and consistency of ESI triage documentation compared with paper-based workflows.

Secondary objectives

- Reduce missing or incomplete triage data fields.
- Improve alignment between recorded triage decisions and rule-based ESI classifications.
- Enhance standardisation of nurse-patient triage communication.

3.5 Intended Impact

This system is designed to support safer and more consistent triage decision-making by reducing documentation variability and improving structured data capture at the point of care. It is intended for use in high-pressure emergency department environments where workload and time constraints can impact triage consistency.

4. CONSTRAINTS & STAKEHOLDERS

4.1 Stakeholder Analysis

The implementation of the proposed digital ESI triage assistant must manage diverse operational priorities across clinical, administrative, and supportive services. In alignment with systematic qualitative review evidence from Hogg et al. (13), who state that successful implementation requires active engagement across multiple distinct stakeholder groups to address individual workflow concerns, Table 1 maps five specific roles at Mercer General to their primary operational priorities. Names used for illustrative stakeholder roles (e.g. Sister Patrice Alleyne) are drawn from the fictional Mercer General case site and are not real individuals.

Table 1. Stakeholder analysis matrix for the Mercer General ED digital triage system.

Stakeholder Role	Representative / Context	Primary Workflow / System Concern
1. Triage Nurse	Sister Patrice Alleyne / Frontline Nursing Staff	Time budget and decision control. Nurses have only 3-5 minutes per patient to collect vital signs, allergies, and complaints. They resist systems that slow them down with extra typing or try to force a clinical decision. Grounded in Gorick et al. (5), the tool must act as a recommendation layer that respects clinical intuition, as environments that pressure staff to maintain flow frequently override the time needed for thorough assessment.
2. ED Consultant	Senior Emergency Physician Leadership	Patient safety and accurate handoffs. Senior doctors worry that a digital tool might under-triage a high-acuity patient, causing dangerous delays. The system must enforce safety-critical, rule-based alerts so that critical cases are never missed during shift handoffs. This risk is highlighted by Agius et al. (1), whose cognitive task analysis shows that deviations from evidence-based protocols can lead to medical errors or inappropriate prioritisation under pressure.
3. Porter / Support	Transport & Logistics Staff	Physical bottlenecks and patient movement. Porters physically move patients to designated areas, such

Stakeholder Role	Representative / Context	Primary Workflow / System Concern
		as acute care beds or fast-track chairs. As shown by De Freitas et al. (2), a lack of porter and attendant availability directly halts patient progression, so the tool must show instant routing updates, so staff do not waste time searching for patient locations or files.
4. Admin Clerk	Front Desk Registration Staff	Data lags and front-desk speed. Clerks are first contact for patients, entering names and arrival modes. They worry that moving from paper to a screen will slow the intake line, especially during surges exceeding 150 daily presentations. Hogg et al. (13) confirm that clerical and technical adoption is heavily blocked when tools disrupt current practice or when organisations lack capacity to support initial launch.
5. Patient Advocate	Patient Liaison / Community Voice	Person-centred care and privacy. This role focuses on reducing patient anxiety and preventing privacy leaks in crowded waiting areas. Phiri et al. (15) establish that patients face extreme frustration and distress when the triage environment lacks privacy or when poor communication leaves them feeling neglected; the tool must help nurses communicate more supportively.

4.2 Candidate Workflow Constraints

Constraint 1: Digitisation Delays and Entry Gaps at Information Points

When using the new touchscreen interface to input vital signs and chief complaints, the system's rule-based safety engine stops progression if mandatory fields are left blank. Sister Patrice points out that during a heavy surge of 110-170 patients, a nurse working a night shift at 03:00 cannot manage a restrictive, long digital form; if the tool exceeds the strict 3-5 minute triage time budget, staff experience cognitive fatigue and skip fields or enter default values to clear the queue. This mechanism is directly supported by Agius et al. (1), who confirm that emergency nurses face constant interruptions and high multitasking demands that increase data entry omissions. Hogg et al. (13) further show that digital health implementations frequently fail when rigid layouts clash with the fast-paced needs of frontline staff, leading to tool abandonment.

Constraint 2: Shared Diagnostic Resources and Limited Computer Access

Frontline staff must share a very limited number of desktop computers at the main desk, meaning the tool cannot require constant screen interaction, while Mercer General's shared laboratory and

X-ray departments mean the triage tool must operate without relying on immediate diagnostic results. This creates a substantial operational bottleneck during peak surges, with turnaround times stretching from 30 minutes to 3 hours because emergency workups compete with regular inpatient hospital demands. This baseline infrastructure constraint aligns with Hogg et al. (13), who highlight that healthcare technology adoption is heavily blocked by unsupportive physical environments and inadequate time. De Freitas et al. (2) similarly show that insufficient material resources force staff into wasteful, time-consuming motion to locate forms across different clinical areas.

Constraint 3: Re-Triage and Waiting-Room Deterioration Are Not Captured by a Single-Decision Tool

Re-triage is an existing part of Mercer practice: a patient initially assigned ESI 4-5 who deteriorates while waiting must be re-categorised, and the triage nurse is expected to visually rescan the waiting room at intervals to catch this. A digital tool that captures only the first triage decision and provides no mechanism for logging a re-assessment would be a regression on current paper practice, not an improvement, because escalation events would no longer have any structured record at all. The mechanism here is one of scope, not just usability: Components 1-3 of the proposed system (Section 3.2) are presently designed around the initial intake decision. Without an explicit re-triage entry point feeding back into the same structured record, deteriorating patients in the fast-track or waiting-room population fall outside the system's data capture, undermining both the patient-safety case and the completeness metric defined in the evaluation plan (Section 3.3).

5. DOOR-TO-DISPOSITION SYSTEM MAP

Figure 1 (Section 3.2) maps the complete door-to-disposition process at Mercer General ED and annotates the five points at which the proposed digital decision-support components (AI 1-AI 5) integrate into the existing clinical pathway, from registration through to structured database capture at disposition.

6. RISK REGISTER

Table 2 presents the full project risk register, spanning AI-technical, operational, ethical, and equity risk categories. Likelihood and impact ratings, mitigations, and measurable signals of

success were derived directly from the literature synthesis in Section 2.1 and are referenced explicitly throughout the proposed solution (Section 3) and risk analysis (Section 7) below.

Table 2. Project risk register for the Mercer General ED digital triage decision-support system (11 risks across AI-technical, operational, ethical, and equity categories).

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
1. Demographic Underrepresentation in Training Data Causing Differential ESI Misclassification by Age and Comorbidity	AI-Technical / Equity	High	High	Source training data from multiple Caribbean ED sites. Conduct subgroup performance audits by age, sex, and presenting complaint prior to deployment. Establish minimum representation thresholds for paediatric and elderly cohorts.	Model sensitivity/specificity metrics within $\pm 5\%$ across demographic subgroups in post-deployment audit.
2. Training-Distribution Shift Between Urban Single-Site Training Data and Multi-Site Caribbean Deployment Contexts	AI-Technical	High	High	Validate the model on held-out data from at least one geographically distinct Caribbean ED before rollout. Implement performance monitoring dashboards that flag drift when ESI distribution deviates $>10\%$ from baseline.	Monthly model performance reports show stable AUC (≥ 0.80) at each deployment site across the first 12 months.
3. High False-Positive Alert Rate Eroding Nurse Vigilance and Increasing Passive Acceptance of AI Output	Operational	High	High	Limit AI recommendations to high-confidence cases (prediction probability ≥ 0.85). Design the interface to present supporting reasoning, not just a score. Conduct quarterly audits of	Override rate remains above 15% (indicating active engagement); clinicians report high perceived usefulness on validated satisfaction surveys.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
				nurse override rates.	
4. Automation Bias Reducing Independent Clinical Reassessment of AI-Assigned Triage Scores	Ethical	Medium	High	Mandate that the system operate in advisory mode only. No triage decision is confirmed without explicit nurse acknowledgement. Include structured training on cognitive biases in AI-assisted decision-making at onboarding.	Prospective audit shows no statistically significant reduction in nurse-initiated ESI reassessments post-deployment.
5. Unconstrained LLM Generation Producing Clinically Unsafe Free-Text Triage Prompts Outside Validated Question Scope	AI-Technical	Medium	High	Apply strict output constraints to the natural-language follow-up module; restrict to closed-ended, validated clinical question sets. Flag any free-text generation outputs for human review before surfacing to the nurse. Run red-teaming sessions on adversarial patient presentations.	Zero instances of clinically unsafe free-text output in 6-month post-launch monitoring log.
6. Infrastructure and Connectivity Failure	Operational	High	Medium	Design the tool to function in offline or degraded-network mode, reverting to a static ESI reference card when connectivity is lost. Implement automatic fallback notification to the charge nurse.	System achieves ≥99% graceful-degradation uptime; paper fallback used in <1% of triage encounters per month.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
7. Clinician Cognitive Overload	Operational	Medium	Medium	Co-design the interface with Mercer ED nurses in usability workshops. Limit AI output to one primary recommendation and one supporting data point visible per screen. Conduct cognitive task analysis before and after deployment.	*NASA-TLX workload scores do not increase significantly from baseline after 90 days of use.
8. Absence of Informed Consent from Patients	Ethical	Medium	High	Develop a clear patient-facing disclosure, available in English and the local Creole, that AI tools are used in triage. Post visible signage at ED reception. Submit consent framework to the Saint Cedric Ministry of Health ethics board prior to deployment.	Ethics board approval obtained; patient survey shows ≥80% recall of AI disclosure at exit interview.
9. Equity Gap: Paediatric and Rural Patients	Equity	High	High	Explicitly test model performance on patients under 18 and on presentations typical of rural referral cases (delayed presentation, atypical vital signs). If subgroup performance is below acceptable threshold, restrict	Model sensitivity for paediatric ESI Level 1-2 cases ≥85% in validation dataset before go-live.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
				deployment to adult urban ED until data can be collected.	
10. Data Privacy and Security	Ethical / Operational	Medium	High	Ensure all patient data processed by the AI module is de-identified at input. Store no identifying information in the model inference pipeline. Conduct a Data Protection Impact Assessment (DPIA) compliant with Saint Cedric health data regulations. Encrypt data in transit and at rest.	Zero reportable data breaches in the first year; DPIA signed off by institutional data protection officer.
11. Regulatory and Liability Ambiguity	Ethical / Operational	Medium	High	Engage the Saint Cedric Ministry of Health and regional health authorities to clarify the liability framework before clinical deployment. Document the advisory-only status of the tool formally in the clinical governance charter.	Formal liability framework agreed and signed by hospital administration and Ministry before go-live.
12. Silent Model Performance Decay as Case-Mix Evolves Beyond the Training Distribution	AI-Technical	Medium	High	Schedule semi-annual model revalidation against incoming Mercer ED data. Define a minimum performance threshold (AUC < 0.75) that triggers	Revalidation reports produced on schedule; no undetected performance decline exceeding 5% AUC over any rolling 6-month window.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
				automatic downgrade to advisory-only mode pending retraining. Assign a named clinical informatics lead as model steward.	

**NASA-TLX (NASA Task Load Index) is a widely used, subjective, multidimensional questionnaire that measures a person's perceived workload while performing a specific task or operating a system.*

7. RISK ANALYSIS

The three risks below are expanded in plain language because each carries a design or governance implication that the register's mitigation column cannot fully convey on its own: the limits of what subgroup testing can guarantee, the limits of interface design against automation bias, and the operational consequence of an offline-first requirement. Each follows the same structure: what the risk means in practice, why it matters for patient safety, what the proposed mitigation does, and an honest assessment of whether that mitigation can be fully defended.

7.1 Risk 1: The Model May Not Know Patients Like Yours

What this means: AI triage tools learn from historical patient data. If that data was collected mostly from hospitals in the United States or Europe, or even from a Caribbean hospital on another island, the model has probably seen very few patients who look like those walking into Mercer General ED, including children, elderly patients, patients who present late due to travel distance, and patients whose comorbidities reflect regional disease patterns such as sickle cell disease, dengue, or poorly controlled hypertension.

Why it matters: a model that under-performs on these groups will not just be less useful, it will actively misclassify them. A child with ESI Level 2 severity could be scored as Level 4 — a patient placed at real risk.

Mitigation: before this tool is deployed at Mercer, it must be tested specifically on the types of patients who come through that ED. If the model cannot achieve equivalent accuracy across age groups and presenting complaints, it should not go live until it can (Table 2, Rows 1 and 8).

Can this be defended? Yes. Subgroup validation is standard practice in medical device regulation and is increasingly required under emerging AI governance frameworks. The signal is measurable: a minimum sensitivity threshold for paediatric high-acuity cases is defined, and deployment does not proceed until it is met.

7.2 Risk 2: Nurses May Stop Trusting Their Own Judgement

What this means: when a confident-looking AI recommendation appears on screen, even experienced clinicians tend to defer to it, even when something feels wrong. This is automation bias, documented across aviation, radiology, and emergency medicine. A nurse who might otherwise reassess a patient showing subtle signs of deterioration may unconsciously accept the AI score and move on.

Why it matters: the triage nurse is the last human safety net before a deteriorating patient is assigned an inappropriate acuity level. If the AI weakens that safety net without replacing it with something equally reliable, the system becomes less safe, not more.

Mitigation: the tool is built and marketed as advisory only (Table 2, Row 4). The nurse confirms every decision, and the interface shows the clinical variables driving the suggestion, not just a score, so the nurse engages with reasoning rather than accepting an output. Regular audits of override rates indicate whether nurses retain independent judgement.

Can this be defended? Partially. Advisory-mode design and reasoning transparency are necessary but not sufficient. Automation bias cannot be fully eliminated through interface design alone. Ongoing training and audit are essential co-mitigations, and this limitation is named openly rather than minimised.

7.3 Risk 3: The System Assumes the Internet Will Be On

What this means: Caribbean EDs operate in infrastructure conditions that differ significantly from the hospitals where most healthcare AI is developed and tested. Power interruptions, intermittent internet connectivity, and limited IT support are operational realities. If the triage tool requires a stable connection to a cloud inference server and that connection drops during a busy Saturday night shift, the system becomes a liability rather than a support.

Why it matters: clinical workflows built around a digital tool do not automatically revert to paper. Nurses who have relied on the prompt for six months may be slower and less confident returning to manual ESI assessment under pressure, and a system failure at peak demand is precisely the wrong time to discover this.

Mitigation: the tool is designed from the start with offline functionality, degrading gracefully to a static, evidence-based ESI reference card that works without a network connection (Table 2, Row 6). The fallback is tested regularly, not only at deployment.

Can this be defended? Yes. Offline-capable clinical tools are technically feasible and this is a design requirement specified from the outset, not retrofitted. Graceful degradation in $\geq 99\%$ of connectivity-failure events is testable in a staging environment before go-live.

8. CONCLUSION

This proposal has set out a nurse-facing, human-in-the-loop digital triage support tool for Mercer General Emergency Department, grounded in a literature base that links operational ED pressure, cognitive load at the point of triage, and the documented safety failures of clinical AI deployed without adequate validation or human-factors design (Sections 1-2). The five-component system architecture (Section 3, Figure 1) deliberately keeps the triage nurse as the final decision-maker at every stage, and each design choice, from advisory-only ESI suggestions to offline degradation, is traceable to a specific, named risk in the project register (Table 2).

The proposal does not claim that these mitigations fully resolve the risks they target. Automation bias is acknowledged as only partially addressable through interface design (Section 7.2), and the evaluation strategy (Section 3.3) is explicit that the available benchmark dataset cannot substitute for Caribbean-representative validation. These limitations are stated directly rather than minimised, in keeping with the proposal's core position: that a triage support tool is only safe to build if its failure modes are named before deployment, not discovered after it.

The 12-week pilot scope defined in Section 1 gives this proposal a falsifiable test: if structured digital data gathering does not measurably reduce missing ESI fields or improve alignment with rule-based ESI classification within that window, the hypothesis is rejected and the design returns to revision rather than proceeding to wider deployment. Constraint 3 (Section 4.2), identified as a

current gap in the system's coverage of re-triage and waiting-room deterioration, is flagged here as the most immediate area for design refinement ahead of pilot implementation. Subject to that refinement and to the subgroup validation thresholds defined in the risk register, the system as proposed offers a credible, governable first step toward structured digital triage capture at Mercer General, and toward the broader goal articulated by ED leadership: that the dataset must be built before the model, and the model must never be allowed to outrun the nurse's judgement it is designed to support.

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